
Analysis U.S Department of State Country Reports (2013-2015) With a Focus on European Union Country

Si Tang Lin, Dilara Özdil, Zahra Eshtiaghi

Data Science and Business Analytics, University of Warsaw, Poland

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ABSTRACT

This study compares multiple advanced topic modeling methodologies to analyze U.S. Department of State Country Reports on Human Rights Practices from 2013-2015. Beginning with traditional Latent Dirichlet Allocation, the project progressively incorporates sophisticated techniques including transformer-based BERTopic modeling, supervised learning with structural guidance, hyperparameter optimization, and advanced evaluation metrics. The analysis reveals systematic thematic differences between EU and non-EU country reports, tracks significant temporal shifts in human rights discourse, and demonstrates the substantial advantages of transformer-based approaches for diplomatic text analysis. Notably, the study uncovers a 3.62% increase in asylum-related discourse within EU reports correlating with the 2015 migrant crisis and achieves near-perfect (99.6%) structural alignment between generated topics and official report sections through supervised learning techniques.

I. INTRODUCTION

Diplomatic documents, especially standardized human rights reports, provide a unique empirical context for computational text analysis. Their generally constant organizational structures coincide with very detailed policy language, making them ideally suited for investigating how different methodological choices impact topic modeling findings. This effort transcends traditional text mining methodologies by methodically applying and comparing several text-analytic techniques to a singular diplomatic corpus, yielding methodological insights pertinent to computational social science research.

In international relations, diplomatic communications are an important tool for figuring out what policies are most important, what strategies are changing, and how both bilateral and multilateral contacts are changing. But the growing number of these documents makes it much harder to analyze them. Qualitative analysis is important for understanding context, but it doesn't work well for quickly finding overall patterns and long-term trends in large groups of texts.

To address this limitation, the research suggests a computational framework that utilizes text mining and Natural Language Processing (NLP) methodologies to enable systematic and reproducible study of diplomatic conversation. This methodology facilitates the detection of underlying thematic structures, temporal variations in sentiment, and patterns of entity prominence and interaction—dimensions that are difficult to consistently ascertain through human coding alone.

The current study focuses on the diplomatic interactions between the United States and the European Union, particularly within a specified historical context. Utilizing various computational techniques on this dataset, the research aims to generate novel, data-driven insights into the architecture and evolution of transatlantic diplomatic interactions, while simultaneously assessing the influence of methodological variations on the resultant substantive analyses.

Research Question

This initiative seeks to address multiple fundamental research inquiries through the use of text mining methodologies. In particular, it looks at how Latent Dirichlet Allocation and other traditional bag-of-words algorithms stack up against transformer-based methods when used on structured diplomatic letters. It also looks into how well unsupervised transformer-based models can pick up on semantic subtleties that go beyond just simple patterns of words that appear together. The study also looks into how supervised learning methods might use document structure to make topics more coherent and easier to understand. Finally, it looks at the systematic

disparities in how human rights are reported in EU and non-EU nations and how these discrepancies change over time.

II. DATA

The information included in this study consists of the State Country Reports on Human Rights Practices, covering 194 countries worldwide from 1999 to 2015. Each observation relates to language taken from a particular article, which means it is a specific part of political debate. The data come from the Harvard Dataverse, which has a consistent and standardized collection of reporting on human rights.

This study focuses on reports from 2013 to 2015, with the goal of tracking changes in sentiment and language style over time and then linking them to important international events that happened during that time period. In total, there are 585 complete reports (195 reports per year).

During this time, reports were continuously made each year on 25 EU member states and about 170 non-EU countries. After preprocessing, the dataset was split into 26,785 text chunks, each with between 120 and 250 words.

A multi-stage preparation pipeline was used on the dataset to make sure that all methods were consistent while still meeting the needs of each modeling strategy. The initial step was to do certain preprocessing processes that are the same for all methods. These included taking off HTML tags and getting raw text, breaking reports up into standard sections (Executive Summary and Sections 1 through 7), and making text chunks with the best lengths between 120 and 250 words. Then, relevant metadata was added to each segment, such as the reporting year, the country of origin, the status of European Union membership, and the specific document section.

During the second phase, preprocessing procedures that were specific to the method were used. For the Latent Dirichlet Allocation (LDA) approach, extra text normalization steps were conducted, such as tokenization, lemmatization, getting rid of stop words, and vocabulary filtering. Conversely, BERTopic-based methods utilized a little further preprocessing to maintain semantic context for transformer-based embeddings.

The dataset has a total of 26,785 text chunks after preprocessing. Of these, 2,757 chunks (about 10.3 percent) came from EU member states. The average country report had roughly 46 sections, which showed that the materials were organized in a regular way. There are eight standard parts to each report. These parts help you compare the reports and do analysis that follows a certain pattern.

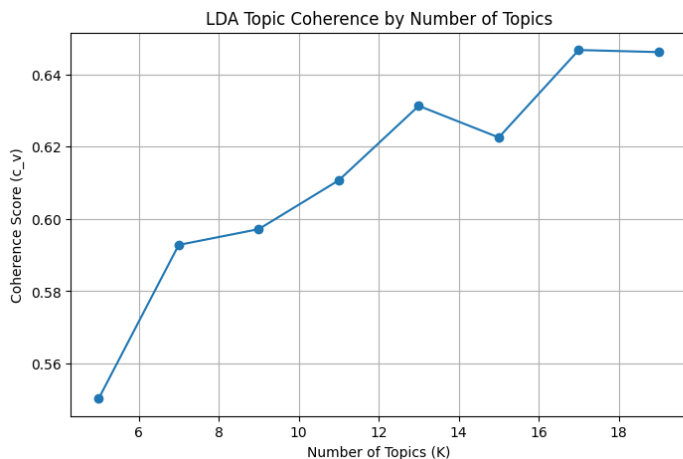
*25 European Union Members: Please note that the present members of the European Union, not the members at the time the reports were produced, define the EU countries.

III. MODELING

The project uses a progressive modeling approach, moving from traditional topic modeling methods to more advanced transformer-based methods. This evolution seeks to methodically assess the influence of increasing model complexity and structural information on topic identification and interpretability in diplomatic writings. The modeling approach includes a baseline Latent Dirichlet Allocation (LDA) model that has been improved using coherence measures, an unsupervised BERTopic model that uses transformer embeddings, and a supervised BERTopic model that uses document structure from report sections. We optimize the hyperparameters for all models, using several validation metrics to make sure the results are strong and can be compared.

3.1 Classic LDA with Coherence Improvement

The first modeling method uses a standard bag-of-words framework, specifically Latent Dirichlet Allocation. The Gensim package is used to implement the model, and a spaCy-based pipeline is used to preprocess the text. After filtering, the final vocabulary has 9,490 different tokens. The minimum document frequency is 10 and the highest document frequency is 50 percent. A coherence sweep, which looks at 5 to 19 topics, determines which model to choose. The best configuration, according to the coherence metric, is 17 topics.



3.2 Unsupervised Transformer-Based BERTopic

The next step uses BERTopic in an unsupervised setting, taking advantage of contextualized embeddings from transformer models. The sentence-transformers/all-MiniLM-L6-v2 model makes sentence embeddings.

We used UMAP to reduce the number of dimensions, using five components and fifteen nearest neighbors. Then we used HDBSCAN to group the data into clusters, with a minimum cluster size of 15. Using class-based TF-IDF (c-TF-IDF), we created topic representations. We then used Maximal Marginal Relevance (MMR) to make the topics' keywords more diverse.

3.3 Supervised BERTopic with Structural Help

The third modeling technique builds on BERTopic by adding supervised information from the structure of the documents. The Executive Summary and Sections 1 through 7 of the report were used as supervisory labels during the embedding and clustering processes. To help with topic generation, section-level metadata was added to the model. The goal was to make the discovered themes fit better with the natural structure of the diplomatic reports. This method tries to find out if clear structural signals can make topics more coherent and easier to understand than methods that don't use any supervision at all.

IV. EVALUATION

4.1 Traditional LDA with Coherence Optimization

The model evaluation used three related criteria: topic coherence, which measured how well the subjects fit together semantically; topic diversity, which measured how much the topics shared words; and dominant topic assignment, which was based on posterior topic probability. The coherence-based optimization produced a final LDA model consisting of 17 topics, derived from a curated vocabulary of 9,490 tokens. The subject diversity score of 0.735 indicates that there is very little overlap across topics, which means that the themes are well-defined and the model is easy to understand. Overall, the topic–word distributions were stable and meaningful in terms of meaning.

The identified subjects align with essential human rights areas commonly included in diplomatic reporting, including judicial processes, law enforcement and security, refugees and asylum seekers, jail conditions, labor rights, freedom of expression, and gender-based violence. Topic

labels were given based on the most likely terms, and then they were looked at again to make sure they made sense and could be understood.

Topic	Difference (EU – Non-EU, %)
Topic 16 (<i>discrimination / person</i>)	+5.56
Topic 11 (<i>disability / person</i>)	+2.71
Topic 2 (<i>refugee / country</i>)	+2.38
Topic 13 (<i>prison / prisoner</i>)	+1.55

Table 1. Topics More Prevalent in EU Countries

Topic	Difference (EU – Non-EU, %)
Topic 4 (<i>election / political</i>)	–0.58
Topic 9 (<i>court / case</i>)	–0.10

Table 2. Topics More Prevalent in Non-EU Countries

A comparative examination of normalized topic distributions demonstrates systematic framing discrepancies between EU and non-EU reports.

EU reports tend to focus more on issues connected to discrimination and minorities, disability rights, refugees and asylum, and jail conditions. Non-EU reports, on the other hand, tend to focus more on judicial and electoral issues. These observed trends suggest substantial geographical disparities in subject emphasis, rather than arbitrary fluctuations in the prevalence of particular topics.

Also, a temporal study from 2013 to 2015 shows that the main themes have changed slowly over time. While issues about the courts and police remained fairly stable, those on refugees and labor became more common, showing that reporting priorities were changing. We used normalized topic proportions to look at these time-based changes and line plots and heatmaps to show them.

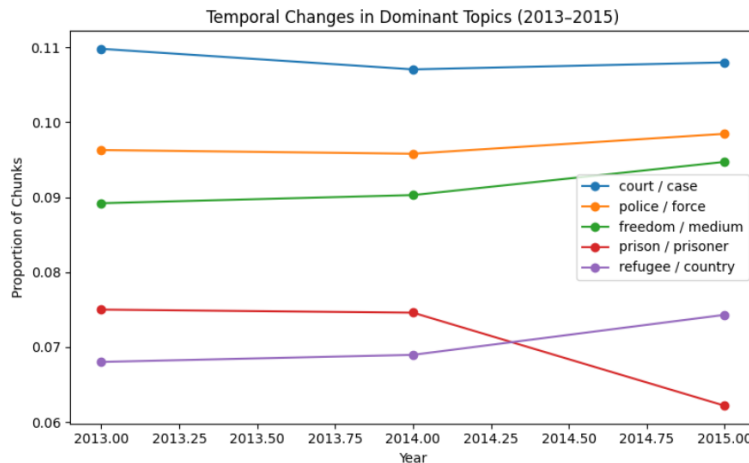


Figure 1. Temporal Changes in Dominant Topics

A historical analysis focused on the European Union highlights a distinct internal realignment in EU reporting. Labor and refugee issues set EU reports apart from non-EU reports. They also show steady development from 2013 to 2015, which shows how important they are becoming in the EU-focused human rights discourse.

The results indicate that the coherence-based LDA model successfully produces a cohesive and comprehensible depiction of human rights discourse in U.S. Department of State country reports. A relatively high topic diversity score of 0.735 indicates that the discovered topics are lexically distinct, which reduces overlap and makes the themes clearer. The predominance of judicial and security-related subjects throughout the whole corpus highlights the centrality of legal institutions and enforcement measures in human rights reporting.

Moreover, persistent disparities between EU and non-EU reporting reveal identifiable regional framing trends. EU reports show a greater attention on concerns like discrimination, disability rights, refugee protection, and jail conditions. This means that there is a stronger focus on social inclusion and rights-based institutional benchmarks. On the other hand, non-EU reports tend to put a little more focus on judicial and electoral processes, which suggests that their reporting is more in line with formal legal and political frameworks.

Temporal study further clarifies a steady transformation in EU reporting from 2013 to 2015. The rise in topics connected to labor and refugees demonstrates that people are becoming more concerned about socioeconomic rights and the challenges of migration. On the other hand, the drop in topics linked to elections and incarceration shows that these themes are becoming less important

over time. These observed patterns collectively demonstrate that subject prevalence is influenced by broader political and social processes, rather than remaining constant.

The strong connection between the main topics and the structure of the reports supports the idea that the LDA results are valid. These observations confirm that traditional LDA, when carefully calibrated and evaluated, can accurately reflect substantial regional and temporal variations in human rights discourse, thereby validating it as a useful and transparent tool for comparative analysis across various countries and time periods.

4.2 Unsupervised Transformer-Based BERTopic

Metric	Value
Initial topics (excluding outliers)	188
Reduced topics (interpretability stage)	30
Coherence score (c_v)	0.458
Topic diversity (top 10 words)	0.50
Model stability (Jaccard similarity)	Mean: 0.86; Median: 0.82
Outlier rate	29.5%

Table 3. Topic Model Performance Metrics

The improved unsupervised model found thirty subjects that could be understood, with a coherence score of 0.458, a topic diversity of 0.50, and a mean stability of 0.86. These metrics indicate that the discovered subjects are semantically coherent, distinct from one another, and constant between iterations. Additionally, around 29.5% of the text segments were designated as outliers, a classification that aided in the segregation of diverse content and ultimately improved the representing quality of the remaining themes.

The identified topics include essential human rights domains, such as labor rights, prison conditions, judicial processes, corruption, press and internet freedom, political participation, gender-based violence, refugee and asylum matters, and issues related to the Roma/Romani minority.

We gave topic labels based on the most likely terms and then checked them by hand to make sure they were easy to understand.

Topic	Δ (EU – Non-EU, %)
Topic 15: Roma / Romani issues	+7.03
Topic 11: Freedom of assembly / association	+3.13
Topic 7: Refugees / asylum	+1.91

Table 4. Topics More Prevalent in EU Countries

Topic	Δ (EU – Non-EU, %)
Topic 10: Police killings	-4.04
Topic 0: Labor rights	-3.05
Topic 1: Prison conditions	-1.91

Table 5. Topics More Prevalent in Non-EU Countries

Normalized topic shares show that there are consistent variances between regions. EU reports often focus on issues that affect the Roma/Romani minority, the right to assemble, and issues that affect refugees and asylum seekers. On the other hand, non-EU papers talk about deaths caused by police, workers' rights, and the conditions in prisons more often. These observed tendencies reflect differing regional emphases; EU reporting prioritizes the protection of minorities and civil freedoms, whereas non-EU reporting lays higher focus on concerns related to conflict and law enforcement.

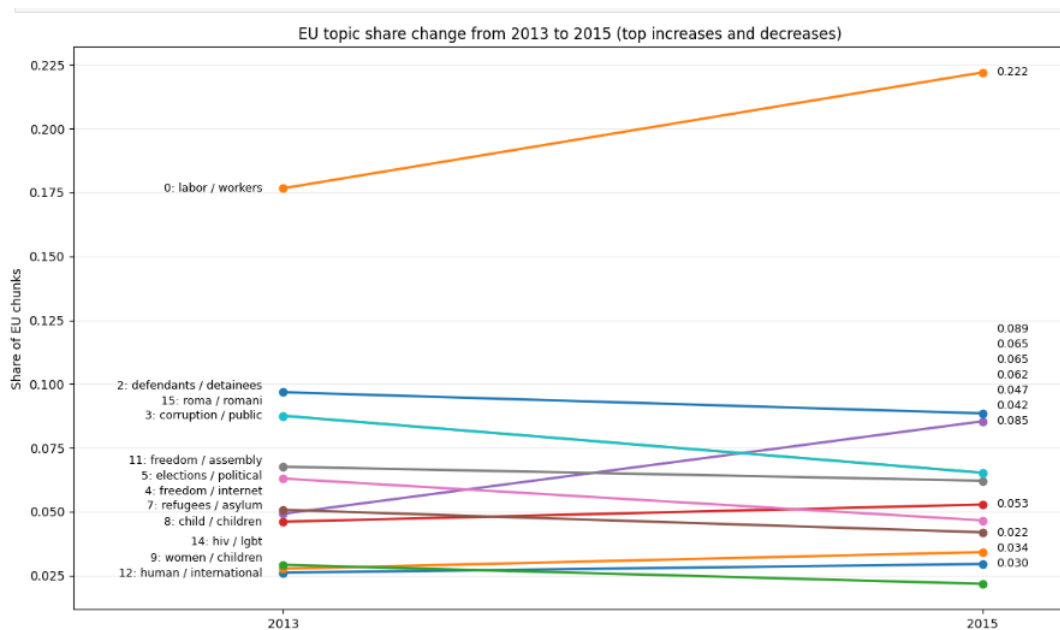


Figure 3. Topic Share Change

Temporal study shows that the way the European Union reports has changed a lot throughout the years. There has been a big rise in the number of articles about labor rights, refugees, and asylum seekers. On the other hand, there has been a little drop in the number of articles about internet freedom, HIV/LGBT rights, and the legal system. The significant increase in asylum-related content is strongly linked to the European migration crisis of 2015. This shows that the model is able to respond to real-world political events.

Even if the model is not supervised, a later comparison with report parts shows that the most important issues and the institutional framework of the reports are very similar. For example, the Worker Rights section (Section 7) is very closely related to topics about workers' rights. On the other hand, topics about prisons are mostly found in the earlier sections. Also, topics about violence against women are mostly found in the Discrimination and Societal Abuses section (Section 6).

The alignment rates are often higher than 95%, which means that the algorithm can find structured thematic content instead of random word groups.

This project shows that using unsupervised BERTopic modeling along with rigorous hyperparameter tweaking and quantitative assessment is a strong and scalable way to study diplomatic texts. The system successfully identifies semantically relevant human rights topics, reveals distinct geographical and temporal trends, and aligns closely with the intrinsic structure of official reports, despite relying solely on unsupervised learning. As a result, the findings confirm the usefulness of unsupervised transformer-based topic modeling as a useful tool for comparing human rights across different regions and time periods.

4.3 Supervised BERTopic with Structural Guidance

Based on the topic modeling performance table below, it is seen that high stability and moderate coherence indicate topics are semantically consistent, reproducible, and distinct, while outlier filtering improves topic clarity by excluding weakly thematic content.

Metric	Value
Final number of topics	30
Coherence (c_v)	0.458

Topic diversity	0.50
Stability (Jaccard overlap)	0.86 mean, 0.82 median
Outlier rate	~29.5%

Table 6. Topic Modeling Performance

The supervised BERTopic model extracted 30 interpretable human rights themes, including:

Topic ID	Interpreted Theme
0	Labor Rights
1	Prison Conditions
2	Judicial Process
4	Press & Internet Freedom
6	Gender-Based Violence
7	Refugees & Asylum
11	Freedom of Assembly
15	Roma / Romani Minority Issues

Table 7. Topics Interpreted by BERT

We figured out the topic labels by looking at the terms that came up the most often, and we made sure they were correct by using structural alignment.

Normalized topic distributions show clear and consistent differences between reports from EU and non-EU countries. EU publications are more likely to talk about Roma/Romani minority issues (+7.03%), freedom of assembly (+3.13%), and refugees and asylum (+1.91%). On the other hand, non-EU papers talk about police killings (−4.04%), labor rights (−3.05%), and jail conditions (−1.91%) more often. This trend shows that EU reports focus more on protecting minorities, asylum processes, and civil freedoms, whereas non-EU reports focus more on human rights issues relating to conflict, security, and enforcement.

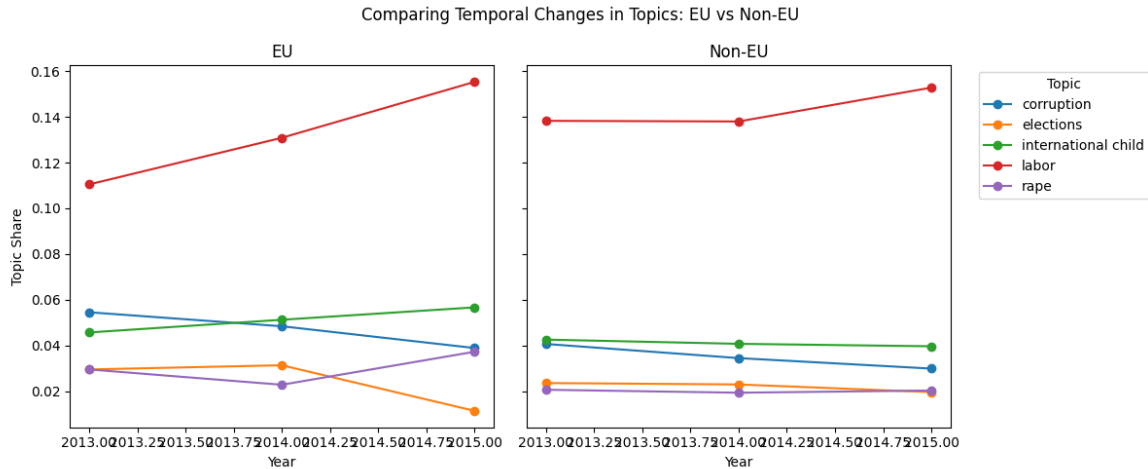


Figure 4. Comparison of Temporal Changes

A temporal examination of EU reports from 2013 to 2015 shows that the themes they focus on have changed a lot.

Labor rights (+4.54%) and refugee and asylum issues (+3.62%) saw the biggest increases, while internet and media freedom (−0.88%), HIV/LGBT rights (−0.74%), and judicial processes (−0.83%) saw the most decreases within the same time period. The considerable rise in asylum-related content is especially remarkable because it closely ties to the European migration crisis of 2015, which shows how well the model responds to important political and humanitarian events in the real world.

Additionally, supervised validation through topic–section alignment provides further corroboration of the model's substantial accuracy. The main topics are very similar to the structure of the original report. For example, labor rights topics are 99.6% similar to Section 7 (Worker Rights), prison conditions are mostly similar to Section 1, and gender-based violence always comes from Section 6 (Discrimination and Societal Abuses).

Alignment rates typically exceed 95% across diverse subjects, so affirming the model's ability to identify structured thematic aspects within the reports, rather than arbitrary word combinations.

These results show that supervised BERTopic modeling is a strong and flexible way to analyze diplomatic messages. By using semantic embeddings, quantitative optimization metrics, and structural supervision, this method gets a lot of semantic depth, strong interpretability by matching

institutional human rights classifications, and policy-relevant insights that show both regional focuses and changes over time. The results substantiate the effectiveness of supervised transformer-based topic modeling as a significant instrument for comparative human rights analysis across various regions and temporal contexts.

V. Summary

There are noticeable differences in how different modeling methods operate when you compare them. Traditional Latent Dirichlet Allocation (LDA) offers distinct topic structures; nevertheless, it fails to encapsulate the complete semantic meaning. In comparison, the unsupervised BERTopic method creates themes about human rights that are more coherent and consistent, giving a better comprehension of the subject in context.

The supervised BERTopic variant enhances interpretability by closely aligning topics with the institutional structure of the reports, thereby confirming that the incorporation of contextual embeddings and structural information enhances topic quality.

The analysis of theme distributions shows that EU reports are more focused on a small number of rights-related topics, such as labor rights, refugee protection, and minority issues. On the other hand, non-EU reports include a wider range of themes, including those connected to enforcement and security. Thus, our results demonstrate significant geographical inequalities in the discourse on human rights, rather than simply errors in reporting.

Temporal study demonstrates that EU reports display greater theme variety in comparison to non-EU reports. From 2013 to 2015, there was a significant increase in coverage of news about refugees and workers, while coverage of stories about the courts and elections decreased. This observation indicates a steady transformation in reporting priorities.

The substantial rise in asylum-related discourse aligns well with the 2015 European migrant crisis, demonstrating the model’s sensitivity to real political occurrences.

A post-hoc analysis of topic-section correspondence shows that the structure is very consistent. For example, labor-related topics are more common in worker rights sections, whereas gender-based violence is more common in discrimination sections. Outlier analysis shows that the text segments that were left out mostly come from narrative-rich or conflict-driven parts. This makes the main theme clearer instead of hurting the model's performance. Stability assessments also show that core subjects stay the same over different model iterations, which strengthens the robustness and trustworthiness of the conclusions.

VI. APPENDIX

Source Code: Available in the GitHub repository: [Github](#) (Including three Jupyter notebooks: [corpus_copy.ipynb](#), [supervised_topic_modeling.ipynb](#), [topicmodeling_Transformer.ipynb](#))

Data Set: [Harvard.edu](#)